

Ema Maria Monteiro Martins

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Education

Universidade do Porto	Sep 2024 - Jul 2026
Master's Degree in Informatics and Computing Engineering	Porto, Portugal
Universidade do Minho	Sep 2022 - Jul 2024
Bachelor's Degree in Informatics Engineering	Braga, Portugal
Universidade do Minho	Sep 2020 - Jul 2024
Bachelor's Degree in Biomedical Engineering	Braga, Portugal

Experience

Mindera	Feb 2026 - Jun 2026
Master Thesis Intern	Porto, Portugal
Developed an internal project aimed at creating an AI agent capable of acting as a second player in two-player video games . Engineered the AI to prioritize human-like behavior and believability during gameplay, optimizing for realistic interactions rather than flawless performance. Integrated computer vision and AI models to process in-game environments in real-time , utilizing NVIDIA NitroGen, YOLO, and Grounding DINO .	

Projects

Escape do Saber Unity	Repo Link
Developed an educational Virtual Reality escape room for the Meta Quest 2 as part of the Virtual and Augmented Reality course. Designed and implemented 3 subject-related educational mini-games focused on color mixing mechanics, anatomy (skeleton assembly), and English language comprehension. Created an interactive ending where users complete the experience by collecting a graduation cap.	
Balloon Popping Robot Competition Unity, ML-Agents	Repo Link
Engineered a MARL competition in Unity using ML-Agents to simulate an autonomous team-based Battle Royale tournament . Applied POCA and Curriculum Learning to train robot agents to navigate asymmetric obstacles and adapt to shrinking arena constraints. Equipped agents with Ray Perception Sensors for real-time obstacle avoidance, offensive tactics, and resource collection.	
Right Under Your Moustache Godot	Repo Link
Co-developed a 2D puzzle-stealth game in Godot as part of the Digital Game Design and Development course. Programmed core gameplay mechanics, including player special abilities, puzzle logic, and enemy detection systems. Designed original visual assets , character models, background elements, and integrated immersive audio and sound effects.	
Races Unity, Vuforia AR	Repo Link
Developed a mobile Augmented Reality (AR) racing game for the Virtual and Augmented Reality course using Vuforia . Programmed core mechanics, including an AI opponent with selectable difficulties, motion-based controls, and a 6-scene UI flow. Designed a strategic phase allowing players to dynamically place obstacles and power-ups on the track using AR markers.	

Skills

Technical Skills	Languages
Game Dev: Godot, Unity	Portuguese: Native
Backend: C, C#, C++, Python, Java, Haskell, .NET	English: Fluent
IA & Machine Learning: Unity ML-Agents, NitroGen, YOLO, Grounding DINO	Spanish: Basic
Frontend: JavaScript, React, Vue.js, HTML/CSS, Pug	
Mobile: Kotlin, React Native, Flutter	
Databases: MongoDB, MySQL, Redis, PostgreSQL, Neo4j	

Leadership & Volunteering

Vice-president of Literatuna – University of Minho	Sep 2023 – Jul 2024
Student Organization Leadership: Coordinated activities and performances of a university musical group.	Braga, Portugal
Erasmus+ Project – ANIMA 2022	May 2022
International Sociocultural Event: International meeting of sociocultural animation.	Portugal / Europe
Volunteer – Extracurricular Classes	Sep 2022 – Mar 2023
Volunteering: Taught extracurricular classes to children aged 6–9, promoting environmental awareness.	Braga, Portugal